Title

Valid Parentheses Using Stack in Java

Objective

To write a Java program to determine whether a string containing parentheses `()`, `{}`, and `[]` is valid using the Stack data structure.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

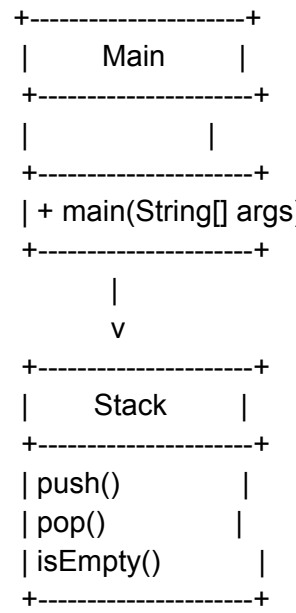
Software Requirements

- Java JDK
- VS Code / Eclipse / IntelliJ IDEA
- Windows/Linux operating system

Algorithm / Logic Description

1. Start the program.
2. Read a string containing parentheses from the user.
3. Create an empty stack to store opening brackets.
4. Traverse each character in the string.
5. If the character is an opening bracket `(`, `{`, or `[`, push it onto the stack.
6. If the character is a closing bracket `)`, `}`, or `]`, check whether the stack is empty.
7. If the stack is empty, mark the string as invalid.
8. Otherwise, pop the top element and check whether it matches the closing bracket.
9. After traversing the string, check whether the stack is empty.
10. If the string is valid and the stack is empty, display `Valid`; otherwise, display `Not valid`.
11. Stop the program.

UML Class Diagram



Source Code

```
import java.util.Scanner;

import java.util.Stack;


public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.print("Enter parentheses: ");

        String s = sc.nextLine();


        Stack<Character> stack = new Stack<>();

        boolean valid = true;
```

```
for (int i = 0; i < s.length(); i++) {
    char ch = s.charAt(i);

    if (ch == '(' || ch == '{' || ch == '[') {
        stack.push(ch);
    }
    else if (ch == ')' || ch == '}' || ch == ']') {
        if (stack.isEmpty()) {
            valid = false;
            break;
        }

        char top = stack.pop();

        if ((ch == ')' && top != '(') ||
            (ch == '}' && top != '{') ||
            (ch == ']' && top != '[')) {
            valid = false;
            break;
        }
    }
}

if (!stack.isEmpty()) {
```

```
            valid = false;
        }

        if (valid) {
            System.out.println("Valid");
        } else {
            System.out.println("Not valid");
        }

        sc.close();
    }
}
```

Compilation and Execution Commands

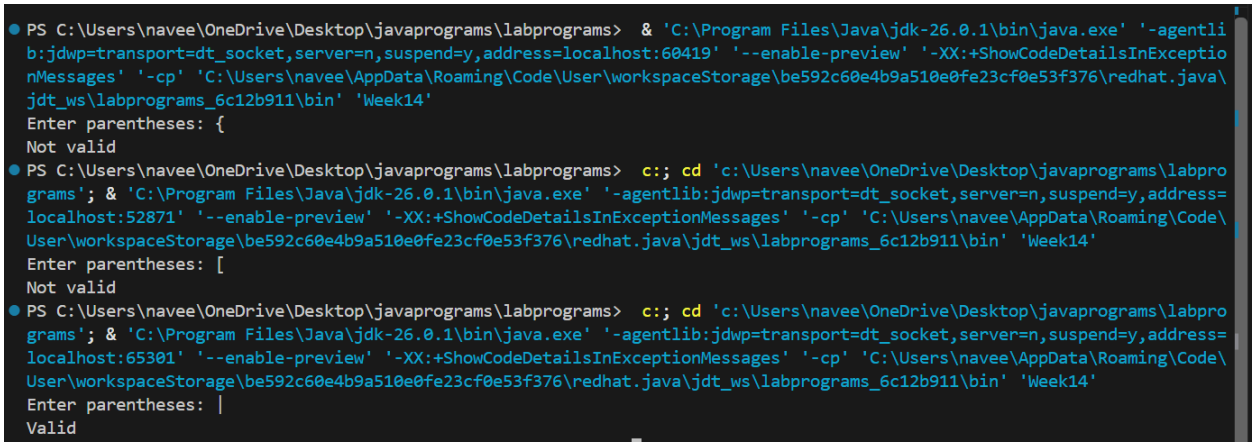
Compile:

javac Main.java

Execute:

java Main

Output Screen



Example 1: Valid parentheses

Enter parentheses: ()
Valid

Example 2: Invalid parentheses

Enter parentheses: ({)}
Not valid

Example 3: Empty string

Enter parentheses:
Valid

Result

The Java program was successfully implemented to check whether a string of parentheses is valid using the Stack data structure.